

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

whether usage of AI for an occupation is primarily a substitute or complement for labor.¹⁵ Table A1 shows example occupations that are in the highest and lowest exposure category for each measure.

Figure 6 shows employment changes by overall prevalence of related Claude queries. The patterns match the findings using the Eloundou et al. (2024) measures closely. Figure 7 likewise shows that the occupations with the highest estimated automation shares have experienced declining employment for the youngest workers.

In contrast, Figure 8 indicates that the occupations with the highest estimated augmentation shares have *not* experienced a similar pattern. Employment changes for young workers are not ordered by augmentation exposure, as the fifth quintile has among the fastest employment growth. The findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.¹⁶

4.4 Fact 4: Employment declines for young, AI-exposed workers remain after conditioning on firm-time shocks

While our results so far are consistent with the hypothesis that generative AI is causing a decline in entry-level employment, there are plausible alternative explanations. One class of explanations is that our patterns are explained by industry- or firm-level shocks correlated with sorting patterns by age and measured AI exposure. For example, one possibility is that young workers with high measured AI exposure are disproportionately likely to sort to firms heavily susceptible to interest rate increases.

We test for a class of such confounders by controlling for a rich set of fixed effects. For each age group, we estimate the Poisson regression

$$\log(E[y_{f,q,t}]) = \sum_{q' \neq 1} \sum_{j \neq -1} \gamma_{q',j} 1\{t = j\} 1\{q' = q\} + \alpha_{f,q} + \beta_{f,t} + \epsilon_{f,q,t} \quad (4.1)$$

¹⁵Figures A2 and A3 show that automation and augmentation results are similar when dropping occupations with low overall Claude usage.

¹⁶Note that occupations in the first two quintiles of the augmentation measure have very low Claude usage overall, with the average occupation in these quintiles comprising 0.01% and 0.09% of conversations, respectively. These occupations have a high share of conversations that are classified as neither automative nor augmentative. In contrast, occupations in the third through fifth quintiles average 0.47%, 0.39%, and 0.33% of Claude conversations. For the automation measure, overall Claude usage increases on average with the automation share, with the lowest exposure group averaging 0.05% of conversations and the highest group averaging 0.73%.

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁷

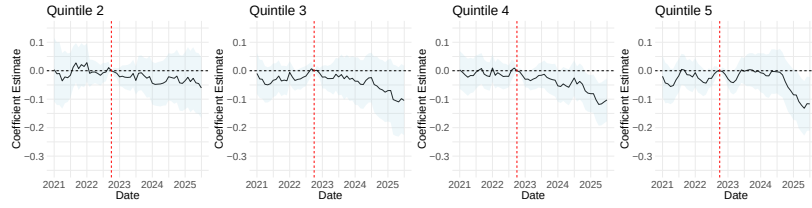
We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁸ Standard errors are clustered by firm.

Results are in Figure 9, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 12 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 3. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

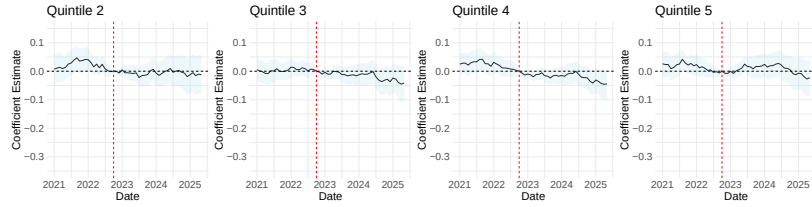
One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁷Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

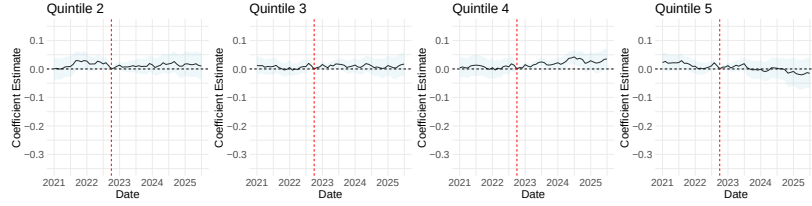
¹⁸Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.



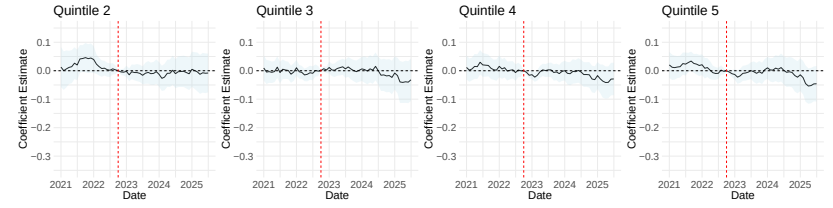
(a) Age 22-25



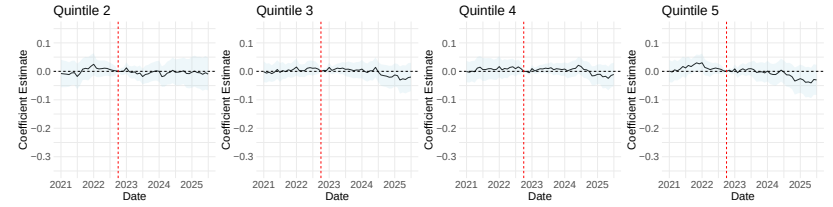
(c) Age 31-34



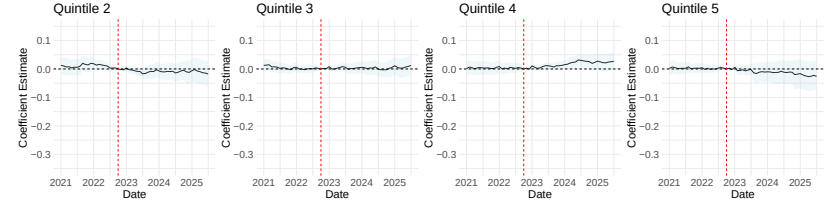
(e) Age 41-49



(b) Age 26-30



(d) Age 35-40



(f) Age 50+

Figure 9: Poisson regression event study estimates for employment changes by age and AI exposure. Estimates are all relative to occupational exposure quintile 1. Exposure quintiles use [Eloundou et al. \(2024\)](#) GPT-4 β measures. Estimates control for firm-time and firm-quintile fixed effects following Equation 4.1. Shaded regions are 95% confidence intervals. Standard errors are clustered by firm.